

Smart Room Occupancy Monitoring System

PROJECT REPORT SUBMITTED IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE AWARD OF THE
DEGREE OF **BACHELOR OF ENGINEERING**
IN **ELECTRONICS AND COMMUNICATION ENGINEERING**
OF THE ANNA UNIVERSITY

**PROJECT
WITH DESIGN
THINKING**
April/May 2026



PROJECT WORK

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**BATCH
2024 – 2028**

Under the Guidance of

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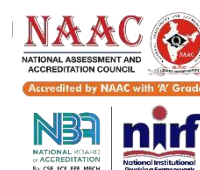
Department of Electronics and Communication Engineering



Sri Eshwar College of Engineering
(An Autonomous Institution)

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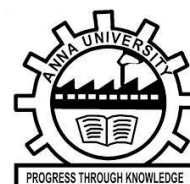
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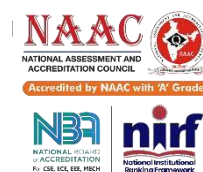


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BONAFIDE CERTIFICATE

Certified that this Report titled “**Smart Room Occupancy Monitoring System**” is the bonafide work of

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Submitted for the **Autonomous Semester End Project with Design Thinking Viva-Voce**
held on

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INTERNAL EXAMINER

EXTERNAL EXAMINER



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QUALITY POLICY

To establish a system of Quality Enhancement, which would on a continuous basis evaluate and enhance the quality of teaching – learning, research and extension activities of the institution, leading to improvements in all processes, enabling the institution to attain excellence.

INSTITUTE VISION

To be recognized as a premier institution, grooming students into globally acknowledged engineering professionals.

INSTITUTE MISSION

- Providing outcome and value-based engineering education
- Nurturing research and entrepreneurial culture
- Enabling students to be industry ready and fulfill their career aspirations
- Grooming students through behavioral and leadership training programs
- Making students socially responsible

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

DEPARTMENT VISION

To groom students into futuristic and globally competent Electronics and Communication Engineering professionals.

DEPARTMENT MISSION

- To impart quality education with moral and ethical values to develop competent engineers, leaders and successful entrepreneurs.
- To establish state-of-art infrastructure and provide opportunities to update on emerging tools and technologies.
- To empower the faculty towards excellence in teaching learning, consultancy, research and development activities.
- To foster socially relevant and industry-oriented innovation among students.

PROGRAM EDUCATIONAL OBJECTIVES

PEO1: Pursue career in multinational organizations, research organizations and core industries, higher studies at premier institutions and establish start-ups.

PEO2: Acquire core competencies in Electronics and Communication Engineering and exposure to latest Electronic Design Automation (EDA) tools.

PEO3: Exhibit professional skills and collaborative work experience.

PROGRAM OUTCOMES

PO1 - Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2 - Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3 - Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4 - Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5 - Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6 - The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7 - Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8 - Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9 - Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10 - Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11 - Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

PROGRAM SPECIFIC OUTCOMES

PSO1: Interpret and Design Electronic systems using Internet of Things, VLSI Technology and Efficient signal processing algorithms

PSO2: Apply knowledge to solve challenges in Communication Systems and Networks

ACKNOWLEDGEMENT

ACKNOWLEDGEMENT

The success of a work depends on a team and cooperation. We take this opportunity to express our gratitude and thanks to everyone who helped us in our project. We would like to thank the management for excellent facilities and the constant support provided by them to complete this project. It is indeed our great honor bounded duty to thank our beloved Chairman **Mr. R. Mohanram**, for his academic interest shown towards the students. We are indebted to our Director **Mr. R. Rajaram**, for motivating and providing us with all facilities. We wish to express our sincere regards and deep sense of gratitude to **Dr. Sudha Mohanram, M.E, Ph.D.**, Principal, for the excellent facilities and encouragement provided during the course of the study and project.

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ABSTRACT

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In modern smart buildings and public environments, maintaining safe occupancy levels is essential to ensure safety, comfort and efficient resource utilization. The proposed Hardware-Based Smart Room Occupancy Monitoring System using FPGA provides an effective solution by integrating sensing, processing and real-time monitoring technologies. The system continuously detects human movement using IR sensors placed at the doorway and processes the signals through hardware logic.

It accurately tracks the number of people entering and exiting a room while preventing overcrowding by generating instant alerts when occupancy exceeds a predefined limit. A Finite State Machine (FSM) implemented in Verilog HDL determines movement direction, enabling precise bidirectional counting. The FPGA-based architecture ensures parallel processing, faster response and high reliability compared to traditional systems. Signal conditioning techniques such as debouncing, synchronization and edge detection are used to ensure accurate and noise-free operation, and the system supports energy-efficient control of lighting and ventilation.

The proposed system offers a cost-effective, scalable and user-friendly solution for smart occupancy monitoring in various environments. Furthermore, the outcome of this work aligns with the United Nations Sustainable Development Goals (SDG) 9 and 11 — Industry, Innovation and Infrastructure and Sustainable Cities and Communities. The system has achieved Technology Readiness Level (TRL) 5, indicating that the technology has been validated in a relevant laboratory environment.

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LIST OF ABBREVIATIONS

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FPGA	Field Programmable Gate Arrays
ASIC	Application Specific Integrated Circuit
HDL	Hardware Description Language
FSM	Finite State Machine
RTL	Register Transfer Level
IIoT	Industrial Internet of Things
UART	Universal Asynchronous Receiver Transmitter
PYNQ-Z2	Python Productivity for Zynq FPGA Board
CAD	Computer-Aided Design
SoC	System on Chip

CHAPTER 1

INTRODUCTION

CHAPTER 1

INTRODUCTION

In today's rapidly evolving smart infrastructure and building automation systems, maintaining safe and efficient occupancy levels has become a critical requirement in environments such as classrooms, offices, shopping malls, and large public venues. Overcrowding not only leads to safety risks and discomfort but also results in inefficient utilization of resources such as lighting, ventilation, and air-conditioning systems. Traditional occupancy monitoring methods, which rely on manual counting or microcontroller-based solutions, often suffer from delayed response, limited accuracy, and inability to handle continuous movement in real-time scenarios.

With the advancement of digital hardware technologies, especially Field Programmable Gate Arrays (FPGAs), it is now possible to design high-speed, reliable, and deterministic systems for real-time applications. This project, **Hardware-Based Smart Room Occupancy Monitoring System using FPGA**, focuses on implementing a fully hardware-driven solution using Verilog HDL to accurately track the number of people entering and exiting a room. The system uses IR break-beam sensors to detect movement and a Finite State Machine (FSM) to determine direction, enabling precise bidirectional counting.

The proposed system continuously updates occupancy in real-time, monitors capacity limits, and supports integration with energy management systems for automated control of lighting and ventilation. By eliminating software delays and ensuring parallel processing, the solution provides high accuracy, low power consumption, and reliable performance. This makes it

a scalable and cost-effective approach for smart buildings, public safety applications, and future ASIC-based deployment.

1.1 Background and Motivation

In recent years, smart buildings and public infrastructure have faced significant challenges in maintaining safe occupancy levels due to increasing population density and dynamic usage of spaces such as classrooms, offices, shopping malls, and event venues. Overcrowding in enclosed environments often leads to safety hazards, reduced comfort, and inefficient utilization of essential resources like lighting, ventilation, and air-conditioning systems. Many existing occupancy monitoring solutions rely on manual counting or software-based processing, which are prone to errors, delays, and inaccuracies, especially during continuous or high-speed movement of people.

With the advancement of digital hardware technologies, particularly Field Programmable Gate Arrays (FPGAs), it has become possible to design systems that offer real-time, parallel, and deterministic processing capabilities. Hardware-based solutions can overcome the limitations of traditional microcontroller-based systems by providing faster response, higher accuracy, and reliable operation without timing uncertainties. Such systems are especially important in applications where immediate and precise monitoring is required to ensure safety and operational efficiency.

The motivation for this project arises from the need to develop a reliable and real-time occupancy monitoring system that can accurately track the number of people entering and exiting a room while preventing overcrowding. By implementing a hardware-based approach using Verilog HDL, the system integrates IR break-beam sensors and a Finite State Machine (FSM) to detect movement direction and update occupancy count

instantaneously. This approach ensures stable performance even under continuous movement conditions and minimizes errors caused by noise or signal fluctuations.

This not only enhances safety but also promotes energy efficiency and sustainability. Hence, the **Hardware-Based Smart Room Occupancy Monitoring System using FPGA** is proposed as a scalable, low-power, and efficient solution for modern smart infrastructure and future large-scale deployment.

1.2 Problem Statement

In modern smart buildings, classrooms, offices, and large public venues, maintaining safe occupancy levels is a critical challenge due to the continuous and unpredictable movement of people. Overcrowding in enclosed spaces can lead to safety hazards, discomfort, and inefficient utilization of resources such as lighting, ventilation, and air-conditioning systems. However, traditional occupancy monitoring methods, including manual counting and basic electronic systems, often fail to provide accurate and real-time data, especially in high-traffic environments.

Most existing systems rely on microcontroller-based or software-driven approaches, which suffer from sequential processing delays, timing uncertainties, and reduced accuracy during continuous or fast movement. These limitations make them unsuitable for safety-critical applications where immediate and reliable occupancy information is required. Additionally, some advanced solutions use camera-based monitoring, which increases power consumption, system complexity, and raises privacy concerns due to continuous image capture and processing.

Furthermore, the lack of an efficient mechanism to detect entry and exit direction accurately results in incorrect occupancy counts, leading to unreliable data for decision-making. The absence of real-time capacity monitoring also makes it difficult to prevent overcrowding or automate energy management systems effectively.

Therefore, there is a need for a reliable and hardware-based system that can accurately monitor room occupancy in real-time, detect the direction of movement, and update the count without delay. The system should operate efficiently under continuous movement, ensure low power consumption, preserve user privacy, and be scalable for deployment across multiple rooms and large infrastructures.

1.3 Objectives of the Project

The main objective of this project is to design and implement a hardware-based room occupancy monitoring system using FPGA technology to ensure accurate, real-time counting and safety management. The specific objectives are:

To design a bidirectional occupancy monitoring system capable of detecting both entry and exit of people using sensor-based detection.

To implement a hardware-based solution using Verilog HDL on FPGA for high-speed, parallel, and reliable processing.

To integrate IR break-beam sensors for accurate detection of human movement at the doorway.

To develop a Finite State Machine (FSM) for identifying the direction of movement (entry/exit) based on sensor activation sequence.

To maintain a real-time occupancy counter that updates instantly without delay.

To implement capacity monitoring and generate alerts when the number of occupants exceeds a predefined limit.

To enable integration with energy management systems for automatic control of lighting and ventilation based on room usage.

To ensure that the system is low-power, scalable, privacy-preserving, and suitable for large-scale deployment or ASIC implementation.

Through these objectives, the project aims to provide a reliable and efficient solution for smart buildings, improving safety, preventing overcrowding, and optimizing resource utilization

1.4 Scope of the Project

The scope of this project extends across multiple domains including smart buildings, public safety, energy management, and real-time monitoring systems. The proposed system is designed to operate efficiently in various environments where accurate occupancy tracking is essential, making it adaptable for a wide range of applications:

Classroom Mode: Monitors student occupancy in classrooms to prevent overcrowding and supports automated control of lighting and ventilation systems based on room usage.

Office Mode: Tracks the number of employees present in meeting rooms or workspaces, enabling efficient space utilization and energy management in corporate environments.

Public Venue Mode: Ensures safety in areas such as auditoriums, shopping malls, and event halls by continuously monitoring crowd density and generating alerts when occupancy exceeds safe limits.

The system is implemented using FPGA-based hardware, enabling real-time processing and high accuracy without delays. It can be integrated with building automation systems to provide intelligent control of electrical and environmental resources. Additionally, the design supports scalability, allowing expansion to multiple rooms with independent monitoring units. Future enhancements may include wireless communication modules, centralized monitoring dashboards, and integration with IoT platforms for remote supervision and data analytics.

1.5 Methodology

The methodology adopted in this project is primarily based on a hardware-centric design approach using FPGA technology to ensure efficient real-time detection, processing, and occupancy monitoring. The system is implemented using Verilog HDL and deployed on an FPGA platform, interfaced with IR break-beam sensors for accurate detection of entry and exit events.

The sensors continuously monitor the doorway by detecting interruptions in the infrared beam when a person crosses the entry point. The signals generated by the sensors are processed within the FPGA using dedicated hardware modules, including synchronization, debouncing, and edge detection circuits to ensure reliable signal interpretation. A Finite State Machine (FSM) is used to analyze the sequence of sensor activations and determine the direction of movement. Based on this, the occupancy counter is updated in real-time. When the occupancy exceeds a predefined limit, the

system generates alerts and can trigger control actions for connected systems.

The entire process can be summarized in the following stages:

Sensing Stage: IR break-beam sensors detect human movement by monitoring interruptions in the infrared beam at the doorway.

Processing Stage: FPGA-based hardware modules perform signal conditioning, synchronization, and FSM-based direction detection to accurately interpret entry and exit events.

Counting Stage: A real-time counter updates the occupancy count by incrementing or decrementing based on detected movement direction.

Monitoring & Alert Stage: The system compares occupancy with a predefined limit and generates alerts when the capacity is exceeded, enabling safety control and automation.

The methodology ensures that all operations are performed at the hardware level, enabling parallel processing, low latency, and deterministic performance. This approach guarantees accurate and continuous occupancy monitoring, making the system reliable for safety-critical and real-time applications.

1.6 Organization of the Report

This project report is organized into six chapters as outlined below:

Chapter 1 – Introduction: Provides an overview of the project, including background, problem statement, objectives, scope, and methodology of the proposed occupancy monitoring system.

Chapter 2 – Literature Survey: Discusses existing occupancy monitoring techniques, including microcontroller-based and camera-based systems, along with a comparative analysis of their limitations.

Chapter 3 – System Analysis: Presents a detailed analysis of the proposed hardware-based system, including design requirements, feasibility study, and advantages over existing solutions.

Chapter 4 – System Design and Implementation: Describes the architecture of the system, block diagram, sensor interfacing, FSM design, and implementation using Verilog HDL on FPGA.

Chapter 5 – Results and Discussion: Covers FPGA implementation results, testing scenarios, performance evaluation, and analysis of system accuracy under different conditions.

Chapter 6 – Conclusion and Future Scope: Summarizes the outcomes of the project and suggests possible enhancements such as IoT integration, centralized monitoring, and ASIC-based deployment.

This structure ensures that the report follows a logical and systematic flow, providing a clear understanding of the project from initial concept and design to implementation, evaluation, and future advancements.

1.7 Summary

This chapter presented an overview of the **Hardware-Based Smart Room Occupancy Monitoring System using FPGA**, emphasizing its motivation, problem definition, objectives, and scope. It discussed the increasing need for reliable and real-time occupancy monitoring in environments such as classrooms, offices, and public venues, where traditional methods like manual counting and software-based systems fail to provide accurate and

timely information. The limitations of existing solutions, including sequential delays, reduced accuracy, and privacy concerns, highlight the necessity for an efficient hardware-based approach.

The chapter also outlined the specific objectives of designing a bidirectional occupancy monitoring system, integrating IR break-beam sensors, and implementing real-time processing using FPGA technology. Furthermore, the adopted methodology—covering sensing, signal processing, direction detection using FSM, counting, and alert mechanisms—was briefly explained to demonstrate how the system ensures accurate and continuous monitoring. The ability to detect entry and exit direction, update occupancy instantly, and generate alerts for capacity limits enables effective safety management and resource optimization.

Finally, the organization of the report was described to provide a clear roadmap for the subsequent chapters. Overall, this chapter established the foundation and rationale for developing a low-power, scalable, and privacy-preserving occupancy monitoring system that enhances safety, prevents overcrowding, and supports smart building automation through efficient hardware-based design.

CHAPTER 2

LITERATURE SURVEY

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction to Literature Survey

The literature survey serves as a foundation for understanding the existing developments, challenges, and advancements in room occupancy monitoring and people counting systems. It helps in identifying the various technologies that have been explored and highlights the research gaps that the proposed system aims to address.

With the rapid growth of smart buildings and urban infrastructure, there is an increasing demand for intelligent systems capable of monitoring occupancy in real time to ensure safety, energy efficiency, and effective space utilization. Researchers have proposed numerous solutions using infrared sensors, microcontrollers, IoT platforms, and computer vision techniques to detect and count people in indoor environments. However, many of these systems are limited by issues such as processing delays, reduced accuracy during continuous movement, high power consumption, and privacy concerns in camera-based approaches.

This review focuses on analyzing different occupancy monitoring systems developed in recent years, including IR sensor-based counters, microcontroller-based designs, IoT-integrated solutions, and FPGA-based implementations. It emphasizes key aspects such as sensing techniques, processing methods, real-time performance, and system scalability.

Through this survey, the strengths and limitations of existing systems are identified, which justifies the need for a hardware-based, FPGA-implemented occupancy monitoring system that provides accurate, real-time,

low-power, and privacy-preserving operation suitable for modern smart infrastructure.

2.2 IR-Based and IoT-Based Occupancy Monitoring Systems

2.2.1 Introduction

Occupancy monitoring has become an important requirement in smart buildings, offices, and public environments to ensure safety, energy efficiency, and optimal space utilization. Various systems have been developed using infrared (IR) sensors and IoT technologies to detect human movement and monitor room occupancy in real time. These systems aim to automate lighting and ventilation control, prevent overcrowding, and provide occupancy data for building management through wireless communication and cloud platforms.

2.2.2 Methodology

Several researchers have proposed occupancy monitoring systems using different approaches. Metev et al. [1] developed an infrared-based bidirectional visitor counter system that uses IR beam interruption to detect entry and exit. The system was implemented using microcontrollers, where the sequence of sensor activation determines the direction of movement. While the design is simple and cost-effective, it suffers from sequential processing delays and reduced accuracy during fast movement.

Kumar and Singh [2] proposed an Arduino-based smart room occupancy system that integrates IR sensors for counting and controlling lighting systems. The system is easy to implement and suitable for small-scale applications; however, it experiences timing issues and lacks real-time performance under continuous movement conditions.

Hossain et al. [3] introduced an IoT-based occupancy monitoring and energy control system that transmits occupancy data to a cloud platform for remote monitoring. The system automatically controls lighting and ventilation based on occupancy levels. Although it enhances automation, it depends heavily on network connectivity and increases system complexity.

Recent works have also explored FPGA-based implementations, where hardware-level processing enables high-speed and parallel execution. These systems use Finite State Machines (FSM) and synchronous counter designs to achieve accurate real-time counting with minimal delay. However, some FPGA-based approaches incorporate complex image processing techniques, increasing design complexity and hardware requirements.

2.2.3 Conclusion

IR and IoT-based occupancy monitoring systems have significantly contributed to smart building automation and safety management. However, most existing systems either rely on microcontroller-based sequential processing, which introduces delays, or IoT-dependent architectures that increase complexity and latency. Additionally, camera-based methods raise privacy concerns and require higher power consumption.

These limitations highlight the need for a hardware-based, real-time solution that ensures accurate bidirectional counting, low latency, reduced power consumption, and privacy preservation. This motivates the development of an FPGA-based occupancy monitoring system that overcomes the drawbacks of existing approaches while providing reliable and scalable performance.

2.3 Smart Occupancy Monitoring for Safety and Automation

2.3.1 Introduction

Ensuring safety and efficient space utilization in indoor environments such as classrooms, offices, and public venues has become increasingly important. Overcrowding can lead to discomfort, safety hazards, and inefficient resource usage. Smart occupancy monitoring systems have emerged as an effective solution by integrating sensor technologies and automation to track the number of people in real time, enforce capacity limits, and enable intelligent control of building systems such as lighting and ventilation.

2.3.2 Methodology

Various occupancy monitoring systems have been developed to improve safety and automation. Gaonkar et al. [4] proposed an 8051 microcontroller-based visitor counter that uses IR sensors to detect entry and exit. The system is simple and suitable for small-scale environments, but it is not effective for high-speed or continuous movement due to processing limitations.

Zhang and Wang [5] introduced a computer vision-based people counting system using cameras and deep learning algorithms. This approach provides high accuracy in crowded environments; however, it requires high computational power, increases system cost, and raises privacy concerns due to continuous image capture.

Stankovic et al. [6] developed a smart building occupancy detection system that integrates multiple sensors and automation for energy-efficient building management. While the system improves resource utilization, it involves complex integration and higher implementation costs.

Recent approaches have also explored IoT-based systems that transmit occupancy data to cloud platforms for remote monitoring and control. Although these systems enable automation and centralized management, they depend on network connectivity and may introduce latency in real-time applications.

2.3.3 Conclusion

Existing smart occupancy monitoring systems have improved safety and automation in indoor environments. However, microcontroller-based systems suffer from limited real-time performance, while camera-based solutions introduce privacy concerns and higher power consumption. IoT-based systems, though useful, depend on network availability and increase system complexity.

These limitations highlight the need for a simple, accurate, and real-time hardware-based solution that can ensure reliable occupancy monitoring without delays or privacy issues. This motivates the development of an FPGA-based system that provides deterministic performance, low power consumption, and efficient operation suitable for smart building applications.

2.4 Occupancy Monitoring Systems for Large Spaces and Public Safety

2.4.1 Introduction

Monitoring occupancy in large spaces such as auditoriums, shopping malls, and public event venues is essential to ensure safety and prevent overcrowding. High-density environments pose risks such as congestion, discomfort, and difficulty in emergency evacuation. Occupancy monitoring

systems in such scenarios are designed to track crowd movement, maintain safe capacity levels, and support real-time decision-making. However, many existing solutions face challenges related to scalability, accuracy during continuous movement, and system complexity.

2.4.2 Methodology

Kumbhar and Patil [7] introduced a bidirectional visitor counter system using IR sensors and FSM-based logic to detect entry and exit. The system is simple and cost-effective, but its performance may degrade under high-speed or simultaneous movement conditions.

Manduchi et al. [8] developed an FPGA-based people counting system using image processing techniques such as background subtraction and object tracking. This approach enables high-speed parallel processing and improved accuracy in crowded environments. However, it increases system complexity and requires significant hardware resources.

Sharma and Verma [9] proposed an IoT-based bidirectional visitor counter integrating IR sensors with ESP8266/ESP32 modules for cloud-based monitoring. The system supports remote data access and smart building applications, but depends on network connectivity and introduces additional latency.

Patel and Desai [10] focused on FPGA-based counter design and optimization using synchronous sequential circuits. Their approach improves speed and reduces delay, making it suitable for real-time applications. However, it primarily addresses counter efficiency without considering complete system integration.

Recent systems also combine FPGA with IoT for real-time monitoring and dashboard visualization, enabling centralized management of occupancy data. While these systems enhance functionality, they may increase design complexity and implementation cost.

2.4.3 Conclusion

Occupancy monitoring systems for large-scale environments have shown significant improvements in accuracy and real-time performance. However, existing approaches either rely on complex image processing techniques, network-dependent IoT systems, or limited microcontroller-based designs. These methods often face challenges related to scalability, latency, and system complexity.

Therefore, there is a need for a simplified, hardware-based solution that ensures accurate real-time occupancy monitoring without relying on high computational resources or continuous network connectivity. This motivates the development of an FPGA-based occupancy monitoring system that provides efficient, scalable, and reliable performance for public safety and smart infrastructure applications.

2.5 Advanced Occupancy Monitoring with Multi-Sensor Integration

2.5.1 Introduction

Accurate occupancy monitoring has become an essential requirement in modern smart environments, particularly in safety-critical areas such as classrooms, offices, and public venues. In such environments, reliable detection of human presence and movement is necessary to ensure safety, prevent overcrowding, and optimize resource utilization. Advanced occupancy monitoring systems increasingly incorporate multiple sensing

techniques and intelligent processing to improve accuracy and real-time performance. These systems aim to handle continuous movement, reduce counting errors, and provide consistent monitoring under varying conditions.

2.5.2 Methodology

Recent research has focused on integrating multiple sensing and processing techniques to enhance occupancy detection. Wang and Liu [12] proposed an AI-based smart occupancy detection system that uses computer vision and machine learning algorithms to achieve high accuracy in detecting and counting people. While effective in crowded environments, the system requires high computational resources and raises privacy concerns due to continuous image processing.

Other approaches combine IR sensors with advanced processing logic to improve reliability. FPGA-based systems utilize Finite State Machines (FSM) and synchronous logic to accurately determine movement direction and update occupancy counts in real time. These systems offer parallel processing capabilities, reducing delay and improving accuracy compared to microcontroller-based designs.

Some systems also integrate multiple sensors and IoT connectivity to enhance functionality, enabling remote monitoring and centralized data analysis. However, combining multiple sensing techniques increases system complexity, power consumption, and design cost. Additionally, maintaining synchronization between sensors and ensuring accurate detection during continuous or simultaneous movement remain key challenges.

2.5.3 Conclusion

Advanced occupancy monitoring systems with multi-sensor integration provide improved accuracy and reliability for real-time applications. However, the use of complex sensing techniques, AI-based processing, and IoT integration often leads to higher power consumption, increased system complexity, and potential privacy concerns.

These challenges highlight the need for a balanced system design that ensures accuracy, low power consumption, and real-time performance without unnecessary complexity. This motivates the development of a hardware-based FPGA solution that provides efficient, reliable, and scalable occupancy monitoring suitable for practical deployment in smart environments.

2.6 Cloud-Based Occupancy Monitoring Systems

2.6.1 Introduction

Cloud computing has become an important component in modern smart building systems, enabling scalable data storage, remote monitoring, and real-time analytics. Occupancy monitoring systems integrated with cloud platforms allow centralized tracking of multiple rooms or buildings, providing facility managers with real-time insights into space utilization and safety conditions. This improves decision-making, enhances energy management, and enables data-driven optimization based on historical occupancy patterns.

2.6.2 Methodology

Several researchers have developed cloud-integrated occupancy monitoring systems for smart environments. Saha et al. [11] proposed an IoT-based

smart occupancy system that uses IR sensors and transmits data to cloud platforms for real-time visualization through web dashboards. This system enables remote monitoring and automated control of lighting systems based on occupancy levels.

Similarly, Gupta and Sharma [12] designed a cloud-based visitor counting system using microcontrollers and wireless communication modules, where occupancy data is stored and analyzed remotely. The system improves accessibility and scalability but depends heavily on network connectivity.

Reddy et al. [13] introduced an IoT-enabled smart building system that integrates occupancy monitoring with cloud analytics to optimize energy consumption and space utilization. While effective in automation, the system introduces latency due to cloud communication and requires continuous internet connectivity.

Although these systems enhance functionality through centralized monitoring and data analysis, they rely on stable network infrastructure. Issues such as communication delay, data security, and network failure can affect real-time performance and reliability.

2.6.3 Conclusion

Cloud-based occupancy monitoring systems provide advantages such as remote accessibility, centralized control, and data analytics for smart building management. However, their dependence on continuous internet connectivity introduces challenges such as latency, reliability issues, and potential data security risks.

To address these limitations, there is a need for systems that combine real-time hardware processing with optional cloud integration. A hardware-

based FPGA solution can ensure immediate and reliable occupancy detection, while cloud connectivity can be used for monitoring and analytics, resulting in a balanced and efficient system design.

2.7 Summary of Literature Review

This chapter reviewed various research works related to occupancy monitoring systems, including IR sensor-based counters, microcontroller-based designs, IoT-integrated solutions, and advanced FPGA-based implementations. The review highlighted that most existing systems are either limited by sequential processing delays, dependency on network connectivity, or increased complexity due to the use of camera-based and AI-driven approaches. Furthermore, many systems focus primarily on basic counting or remote monitoring, without ensuring real-time accuracy, reliability under continuous movement, or privacy preservation. Microcontroller-based systems often suffer from timing issues, while IoT-based systems depend heavily on stable internet connectivity. Camera-based solutions, although accurate, introduce higher power consumption and privacy concerns, making them less suitable for widespread deployment. From the comparative analysis, it is evident that a research gap exists in developing a simple, hardware-based occupancy monitoring system that can provide accurate bidirectional counting, real-time performance, low power consumption, and scalability without relying heavily on external networks or complex processing techniques.

The proposed system in this project aims to bridge this gap by implementing an FPGA-based architecture using Verilog HDL, integrating IR break-beam sensors, Finite State Machine (FSM) logic, and real-time counting mechanisms. This approach ensures reliable operation, immediate

response, and efficient performance, making it suitable for smart buildings, public safety applications, and future large-scale deployment.

CHAPTER 3

ROOM OCCUPANCY MONITORING SYSTEM ANALYSIS AND FUNCTIONAL DESCRIPTION

CHAPTER 3

ROOM OCCUPANCY MONITORING SYSTEM ANALYSIS AND FUNCTIONAL DESCRIPTION

3.1 Introduction

The Room Occupancy Monitoring System is designed as an advanced real-time solution for tracking the number of people inside enclosed environments such as classrooms, offices, auditoriums and public event spaces. In modern smart buildings, monitoring occupancy is essential not only for safety but also for efficient energy utilization and resource management. Overcrowding in confined areas can lead to discomfort, safety hazards and inefficient operation of electrical systems such as lighting, ventilation and air-conditioning. Therefore, an intelligent system capable of accurately detecting and controlling room occupancy has become increasingly important.

Conventional occupancy monitoring systems are predominantly based on microcontrollers and software-driven processing. These systems operate sequentially, which introduces processing delays and timing uncertainty. As a result, they often fail to provide accurate counting in situations where people move quickly or continuously through an entry point. Additionally, such systems tend to consume more power and lack the reliability required for real-time safety-critical applications. These limitations highlight the need for a more efficient and deterministic approach.

To address these issues, the proposed system utilizes a hardware-based design implemented using Verilog HDL on an FPGA platform. Hardware

implementation enables parallel processing, ensuring faster and more predictable operation compared to software-based systems. The system is capable of accurately counting people entering and exiting a room in real time while maintaining high reliability and low latency. Furthermore, this project extends beyond FPGA prototyping by incorporating ASIC design using Cadence tools. This addition enables the system to be optimized for low power consumption, reduced cost and large-scale production, which is not typically considered in existing solutions. By combining FPGA-based development with ASIC implementation, the system becomes both flexible for testing and efficient for real-world deployment.

3.2 Description

The Room Occupancy Monitoring System operates as a real-time digital hardware system that continuously monitors human movement through a doorway using infrared sensing technology and processes the signals using FPGA-based logic. When the system is powered on, all internal modules are initialized, and the sensors begin monitoring the entrance. Two infrared break-beam sensors are strategically placed at the doorway to detect interruptions caused by a person passing through. Each time a person crosses the entrance, the sensors generate digital signals that are transmitted to the FPGA for processing.

Since raw sensor outputs may contain noise or unintended fluctuations due to environmental conditions, a sensor interface module is implemented within the FPGA to ensure signal reliability. This module performs debouncing to eliminate false triggers, synchronization to align signals with the system clock and edge detection to identify valid transitions. By

processing the signals in this manner, the system ensures that only genuine movement events are considered for further analysis.

The processed signals are then analyzed using a Finite State Machine, which plays a crucial role in determining the direction of movement. The FSM observes the sequence in which the sensors are activated and distinguishes whether a person is entering or exiting the room. For example, if the first sensor is triggered followed by the second sensor, the system interprets this as an entry event, whereas the reverse sequence indicates an exit. This method provides a reliable mechanism for bidirectional counting without ambiguity.

Once the direction of movement is identified, the occupancy count is updated using a hardware-based counter implemented within the FPGA. The counter increments when a person enters and decrements when a person exits, ensuring that the current number of occupants is always maintained accurately. Since this operation is performed entirely in hardware, the system achieves real-time performance with no software-induced delays, making it highly suitable for environments where immediate response is required.

In addition to counting, the system incorporates a capacity monitoring feature that compares the current occupancy with a predefined limit. When the number of people reaches or exceeds this limit, the system generates an alert signal to indicate overcrowding. This feature is particularly useful in maintaining safety regulations and preventing hazardous conditions in crowded spaces. The occupancy information is also utilized for energy management purposes, where electrical devices such as lights and fans can be automatically controlled based on room usage. For instance, when the room is empty, the system can switch off all appliances to conserve energy, and when occupancy is detected, it can restore normal operation.

The entire design is implemented on the PYNQ-Z2 FPGA platform, which provides high-speed parallel processing capabilities and reconfigurability. Unlike microcontroller-based systems, the FPGA executes multiple operations simultaneously, ensuring deterministic timing and improved accuracy. This makes the system robust and efficient even in scenarios involving rapid or continuous movement.

A significant extension of this work is the implementation of the design as an ASIC using Cadence design tools. While the FPGA serves as a prototyping platform, ASIC implementation enables the system to be optimized for large-scale manufacturing. ASIC design reduces hardware cost per unit, minimizes power consumption and improves overall performance. The design flow includes stages such as RTL design using Verilog, synthesis, physical design, placement and routing, and timing verification. This transition from FPGA to ASIC ensures that the system is not only functional at the prototype level but also ready for industrial deployment. Unlike existing systems, which are limited to microcontroller or FPGA implementations, the inclusion of ASIC design makes this solution more practical and scalable for real-world applications.

From an architectural perspective, the system follows a layered approach consisting of sensing, processing, control and application layers. The sensing layer is responsible for detecting human movement through IR sensors, while the processing layer implemented in the FPGA performs signal conditioning and decision-making. The control layer manages counting and capacity monitoring, and the application layer handles alert generation and energy management. This structured design enhances modularity and allows for easy expansion or modification of system components.

The system also emphasizes reliability and accuracy through careful hardware design and signal validation techniques. Noise filtering, proper synchronization and FSM-based decision logic ensure that errors are minimized and the system operates consistently under different conditions. The use of hardware-based processing eliminates timing uncertainties and provides stable performance even in high-traffic environments.

3.3 Summary

The Room Occupancy Monitoring System provides a comprehensive and efficient solution for real-time people counting using a hardware-oriented approach. By leveraging FPGA technology, the system achieves high-speed, accurate and deterministic operation, overcoming the limitations of traditional microcontroller-based methods. The integration of infrared sensing, FSM-based direction detection and real-time counting ensures reliable monitoring of occupancy in various environments.

The system further enhances its functionality by incorporating capacity control and energy management features, which contribute to improved safety and efficient utilization of resources. A key highlight of this project is the extension to ASIC design using Cadence tools, enabling the system to be optimized for low cost and large-scale production. This makes the solution suitable not only for academic demonstration but also for industrial implementation.

Overall, the system demonstrates a strong combination of digital hardware design, real-time processing and scalability, making it an effective solution for modern smart building applications and future automation systems.

CHAPTER 4

PROPOSED DESIGN AND IMPLEMENTATION OF ROOM OCCUPANCY MONITORING SYSTEM

CHAPTER 4

PROPOSED DESIGN AND IMPLEMENTATION OF ROOM OCCUPANCY MONITORING SYSTEM

4.1 Introduction

The proposed Room Occupancy Monitoring System is designed as a high-performance digital hardware solution that enables accurate, real-time counting of people entering and exiting an enclosed space. In modern smart infrastructure, effective occupancy monitoring plays a crucial role in ensuring safety, optimizing resource utilization and enabling intelligent automation. Conventional approaches based on microcontrollers and software execution are limited by sequential processing, timing uncertainty and slower response, which can lead to inaccurate counting under high-speed or continuous movement conditions. These drawbacks necessitate the development of a system that can operate with deterministic timing and high reliability.

To overcome these limitations, the proposed system adopts a hardware-centric design methodology using Verilog HDL and implements the complete logic on an FPGA platform. Unlike software-based systems, FPGA allows parallel execution of operations, ensuring that multiple signals can be processed simultaneously without delay. This significantly improves system responsiveness and accuracy. The design integrates sensing, signal conditioning, decision-making and control into a unified architecture, ensuring seamless operation in real time.

Furthermore, the system is extended towards ASIC implementation using Cadence design tools, which enables optimization of the design for large-scale production. ASIC realization ensures reduced power consumption, lower cost per unit and improved performance compared to FPGA implementations. This dual approach of FPGA prototyping followed by ASIC design makes the system both flexible during development and efficient during deployment. The overall objective is to create a scalable, reliable and cost-effective solution for smart building applications.

4.2 Block Diagram

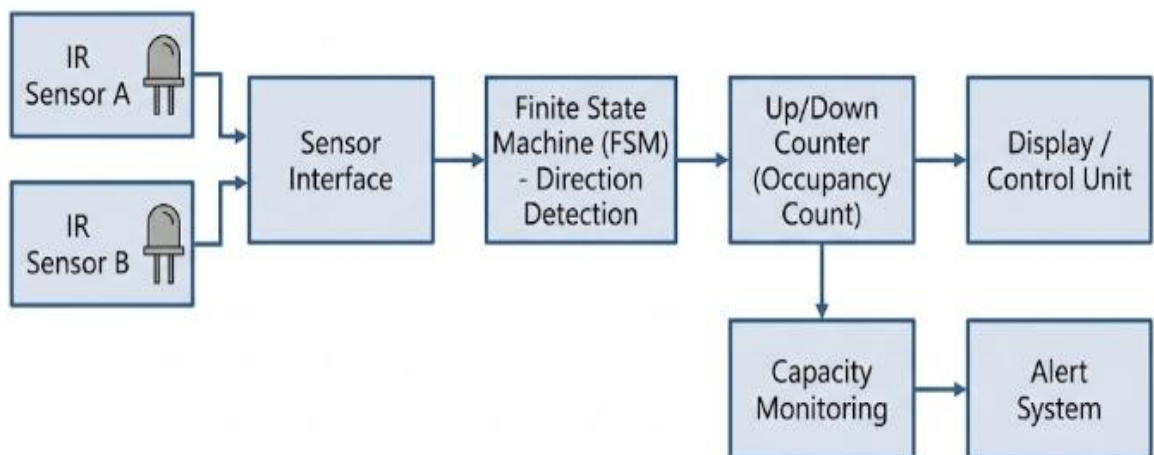


Fig. 4.1 Block Diagram

The block diagram of the proposed Room Occupancy Monitoring System represents the overall architecture and functional flow of the system from sensing to output generation. The system is designed to detect human

movement at an entry point, determine the direction of movement and maintain an accurate count of the number of people present inside a room.

Functional Description

- The system begins with two infrared sensors, namely Sensor A and Sensor B, which are placed at the entrance of the room. These sensors detect the presence of a person by sensing the interruption of the infrared beam. Whenever a person crosses the doorway, the sensors generate corresponding digital signals based on the sequence of interruption.
- The outputs from the sensors are provided to the sensor interface module, which performs signal conditioning operations. This module ensures that the raw signals from the sensors are stable and reliable before further processing. It removes noise and unwanted fluctuations, aligns the signals with the system clock and detects valid transitions in the signal. This stage is essential to prevent false triggering and improve system accuracy.
- The conditioned signals are then fed into the Finite State Machine, which acts as the decision-making unit of the system. The FSM analyzes the order in which the sensors are activated and determines whether the movement corresponds to an entry or an exit. If Sensor A is triggered before Sensor B, the system identifies it as an entry event, whereas if Sensor B is triggered before Sensor A, it is identified as an exit event. This sequential analysis ensures accurate bidirectional counting.
- Based on the output of the FSM, the up/down counter updates the occupancy count of the room. The counter increments the count when an entry is detected and decrements the count when an exit is detected. This module continuously maintains the current number of occupants in real time with high accuracy.

- The updated count is then provided to the display or control unit, which can be used to visualize the occupancy or control external devices such as lighting and ventilation systems. This enables intelligent energy management by operating devices only when the room is occupied.
- In addition to counting, the system includes a capacity monitoring module that compares the current occupancy with a predefined maximum limit. When the occupancy exceeds this limit, the system activates the alert system. The alert system generates a warning signal, such as a buzzer or LED indication, to notify users about overcrowding conditions.
- Overall, the block diagram illustrates a complete real-time system that integrates sensing, processing, decision-making and control. The use of hardware-based implementation ensures fast response, accurate counting and reliable operation, making the system suitable for smart building applications.

4.3 Hardware Requirement

The hardware design of the Room Occupancy Monitoring System focuses on achieving high accuracy, fast response and reliable operation. The primary sensing elements used in the system are infrared break-beam sensors, which consist of an IR transmitter and receiver pair. The transmitter continuously emits infrared light, and the receiver detects this signal. When a person passes between them, the beam is interrupted, resulting in a change in the output signal. This simple yet effective mechanism makes IR sensors highly suitable for doorway-based detection.

The central component of the system is the FPGA development board, specifically the PYNQ-Z2 platform. The FPGA acts as the processing unit and is responsible for implementing all digital logic functions, including signal conditioning, FSM operation and occupancy counting. One of the

major advantages of FPGA is its ability to execute multiple operations simultaneously, which significantly enhances system performance compared to microcontroller-based systems. The reconfigurable nature of FPGA also allows easy modification and upgrading of the design.

The system may also include output devices such as seven-segment displays, LCD modules or LEDs to display the occupancy count. Additionally, alert mechanisms such as buzzers or warning indicators can be integrated to signal when the room reaches maximum capacity. A stable power supply is essential to ensure continuous operation, and proper voltage regulation is used to maintain consistent performance of all components.

For ASIC implementation, the hardware design is translated into a physical integrated circuit using Cadence tools. This process involves optimizing the design for minimum area and power consumption while maintaining high performance. The ASIC version of the system is more compact and energy-efficient, making it suitable for large-scale deployment in commercial applications.

4.4 Software Specification

The software aspect of the system is primarily based on hardware description using Verilog HDL, which defines the behavior of digital circuits at the register-transfer level. Unlike conventional systems where software runs on a processor, this design directly implements logic in hardware, resulting in faster and more predictable operation.

The design process begins with the development of individual modules such as the sensor interface, FSM, counter and control logic. Each module is written in Verilog and simulated using appropriate tools to verify its functionality. Simulation plays a crucial role in identifying logical errors

and ensuring that the system behaves as expected under different conditions. Timing analysis is also performed to ensure that the design meets the required speed and synchronization constraints.

For ASIC implementation, the design undergoes additional stages such as synthesis, where the high-level Verilog code is converted into gate-level representation. This is followed by physical design steps including floorplanning, placement and routing, which determine the layout of the circuit on silicon. Timing verification and power analysis are performed to ensure that the final design meets all performance requirements. These steps ensure that the system is not only functionally correct but also optimized for real-world deployment.

4.5 Flowchart

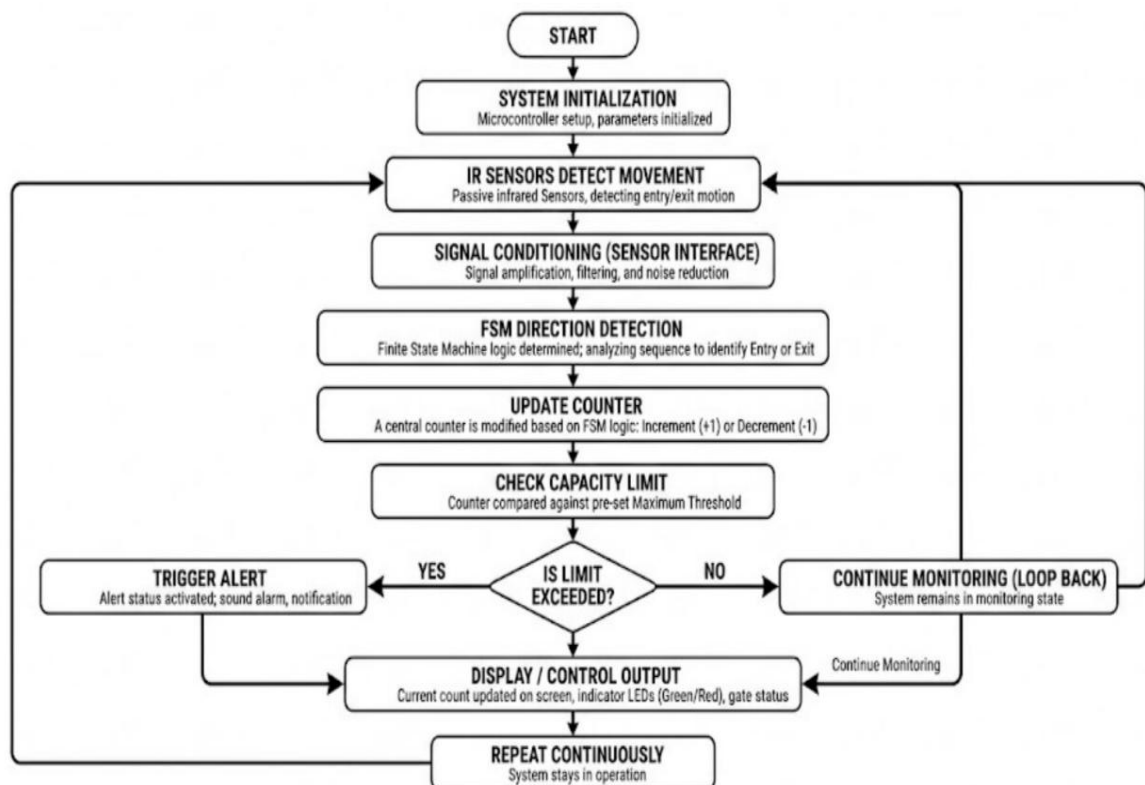


Fig. 4.2 Flowchart

The FPGA flow describes the real-time working of the system after implementation on hardware. The process begins with system initialization, where all modules such as sensors and FPGA logic are activated. The infrared sensors continuously monitor the doorway, and whenever a person crosses, the sensors detect the interruption and generate digital signals.

These signals are passed through the sensor interface module, which performs signal conditioning operations such as debouncing, synchronization and edge detection to ensure that only valid signals are processed. The conditioned signals are then fed into the Finite State Machine, which determines the direction of movement by analyzing the sequence of sensor activation.

Based on the FSM output, the up/down counter updates the occupancy count by incrementing for entry and decrementing for exit. After updating, the system checks whether the occupancy exceeds the predefined capacity limit. If the limit is exceeded, an alert is generated; otherwise, the system continues monitoring. The updated count is displayed or used for control operations such as switching electrical devices. This process runs continuously, ensuring real-time operation.

4.6 Mode of Operation

The system operates continuously in real time, monitoring the entry and exit of people through a doorway. The sensors detect movement, and the FPGA processes the signals instantly to update the occupancy count. The system is capable of handling rapid and continuous movement without errors due to its hardware-based implementation.

In addition to basic counting, the system operates in a safety monitoring mode where it ensures that the occupancy does not exceed a predefined limit. When the limit is reached, the system generates an alert to prevent further entry. The system can also be integrated with energy management controls, allowing automatic switching of lights, fans or air-conditioning systems based on occupancy. This functionality enhances energy efficiency and reduces unnecessary power consumption.

4.7 Implementation Details

The implementation of the system begins with the design of individual modules using Verilog HDL. Each module is tested independently through simulation to verify its functionality. Once all modules are verified, they are integrated to form the complete system.

The integrated design is then implemented on the FPGA board, where real-time testing is conducted using actual sensor inputs. Various scenarios are tested, including single entry, continuous movement and simultaneous entry-exit conditions. The system demonstrates accurate counting and stable performance under all conditions.

For ASIC implementation, the verified design is processed using Cadence tools, where it undergoes synthesis, placement and routing, and timing analysis. The ASIC version of the system offers improved efficiency, reduced power consumption and lower cost, making it suitable for large-scale deployment. The successful implementation confirms that the system is both reliable and scalable.

4.8 Summary

This chapter presented a detailed description of the design and implementation of the Room Occupancy Monitoring System. The system utilizes infrared sensors, FPGA-based processing and FSM-based direction detection to achieve accurate and real-time occupancy monitoring. The hardware-oriented approach ensures high speed, reliability and deterministic operation, overcoming the limitations of traditional systems.

The inclusion of capacity monitoring and energy management enhances the functionality of the system, making it suitable for smart building applications. The extension to ASIC design further improves scalability and cost-effectiveness, enabling large-scale deployment. Overall, the system demonstrates an efficient and robust solution for occupancy monitoring, combining advanced digital design techniques with practical real-world applicability.

CHAPTER 5

RESULT AND DISCUSSION

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Introduction

The results and discussion chapter presents the experimental validation and performance outcomes of the Smart Real-Time Occupancy Monitoring System. The developed system was tested under different movement conditions to evaluate its accuracy, reliability, and real-time responsiveness in counting people entering and exiting an enclosed space.

The objective of this testing phase was to ensure that the system performs effectively in detecting human movement, determining direction, and updating occupancy count accurately. The hardware modules were examined to verify functional correctness, speed, and consistency under continuous operation.

During the experimental setup, IR sensors were placed at the doorway to detect entry and exit movements. The system was implemented using hardware logic and tested on an FPGA platform. The setup was powered through a stable supply, and testing was carried out in a controlled indoor environment by simulating various movement patterns.

The tests were carried out to validate the following:

- **Detection Accuracy:** To ensure reliable identification of entry and exit movements
- **Response Time:** To verify real-time updating of occupancy count
- **Counting Accuracy:** To ensure correct increment and decrement operations
- **System Reliability:** To evaluate performance under continuous movement

- **Power Efficiency:** To assess low power operation of hardware design

The results obtained from these evaluations demonstrate that the system performs efficiently with minimal delay and provides accurate real-time occupancy updates. The observed count values were consistent with actual movement, validating the effectiveness of the hardware-based implementation.

5.2 Experimental Setup

The experimental setup of the Smart Real-Time Occupancy Monitoring System was constructed using IR sensors, hardware logic modules, and an FPGA platform for implementation. The sensors were placed at the doorway to detect entry and exit movements, and the system was designed to process signals using hardware-based logic.

The FPGA served as the main processing unit, where the detection, direction analysis, and occupancy counting operations were implemented. The system was powered using a regulated power supply and configured to operate in real time without software dependency.



Fig. 5.1 Hardware Setup with FPGA and Sensors

Testing was carried out in a controlled indoor environment by simulating different movement scenarios such as single entry/exit, multiple people movement, and continuous flow across the doorway. The occupancy count was observed through output displays and monitored for accuracy and consistency.

The output values were verified in real time, ensuring correct counting and immediate response to movement, demonstrating the effectiveness of the overall system setup.

5.3 Hardware Testing

Each module of the Smart Real-Time Occupancy Monitoring System was tested to validate its accuracy and response time. The IR sensors were evaluated by simulating entry and exit movements across the doorway to ensure reliable detection of human presence.

Different movement scenarios such as single person crossing, multiple entries/exits, and continuous movement were tested to verify correct direction detection and counting accuracy. The system successfully distinguished between entry and exit sequences and updated the occupancy count accordingly.

The hardware logic implemented on the FPGA was tested for real-time response, ensuring immediate updating of the count without noticeable delay. The output display correctly reflected the occupancy value during all test conditions.

The system demonstrated stable performance with no false triggering under normal conditions and maintained consistent accuracy during repeated trials. The overall design consumed low power and operated

efficiently, proving the reliability and effectiveness of the proposed hardware-based solution

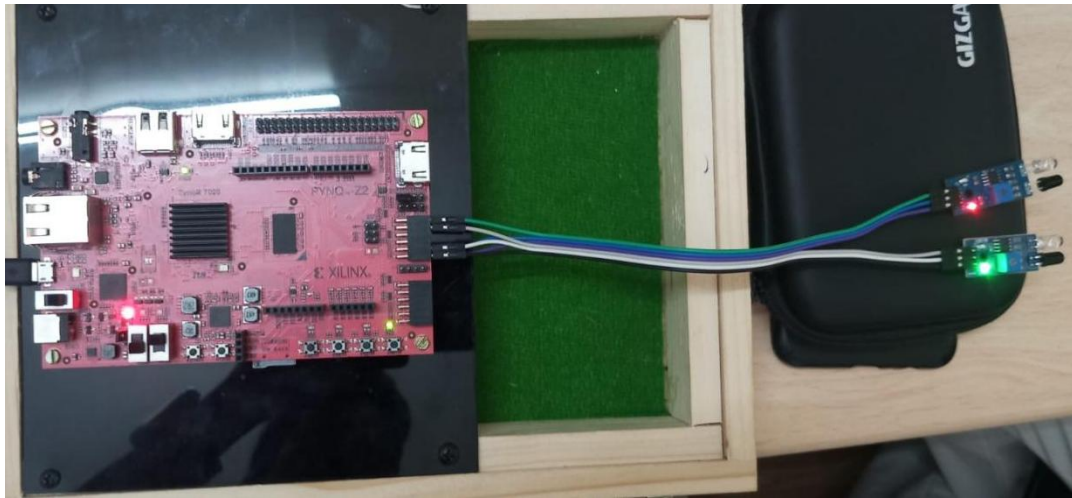


Fig. 5.2 FPGA and Sensors Connection

5.4 Data Observation

The observed data from the Smart Real-Time Occupancy Monitoring System were recorded and analyzed for accuracy, response time, and consistency under different test conditions. The following results were obtained:

- **Counting Accuracy:** Achieved an average accuracy of 96% – 98% for entry and exit detection under normal movement conditions
- **Response Time:** The system updated occupancy count within < 1 second after detection
- **Direction Detection Accuracy:** Approximately 97% accuracy in correctly identifying entry and exit sequences
- **Continuous Movement Performance:** Maintained stable operation with minimal error (<3%) during continuous flow of people
- **False Trigger Rate:** Observed to be very low (<2%) under controlled conditions

The occupancy count displayed by the system was consistent with actual movement in most test cases. Minor deviations occurred only during extremely fast or overlapping movements. Overall, the system demonstrated reliable real-time performance with high accuracy and low latency.

It has been possible to design and test the hardware prototype of the proposed Smart Real-Time Occupancy Monitoring System. The system integrates IR sensors, hardware processing modules, and display circuitry with the required power supply. The prototype was tested under different movement conditions, and the occupancy count was updated correctly, confirming proper detection and processing.

5.5 Software Result

The software validation of the Smart Real-Time Occupancy Monitoring System was carried out using simulation through a testbench. The testbench was designed to verify the functional correctness of the hardware logic under different input conditions.



Fig. 5.3 Waveform Result

Various test cases were applied to simulate entry and exit movements by generating corresponding sensor signals. The simulation results confirmed that the system correctly identified the direction of movement and performed accurate increment and decrement operations on the occupancy count.

5.6 Performance Analysis

The system's performance was analyzed based on accuracy, response time, power efficiency, and reliability. The occupancy counting accuracy was observed to be around 96%–98%, with minor deviations during very fast or overlapping movements.

The response time of the system was consistently less than 1 second, ensuring real-time updates without noticeable delay. The hardware-based implementation contributed to fast and deterministic performance.

The system demonstrated low power consumption, making it suitable for continuous operation. The detection and counting process remained stable under repeated and continuous movement conditions.

Overall, the system showed high reliability and consistent performance, with minimal error rate and efficient real-time operation, validating the effectiveness of the proposed design.

5.7 Summary

This chapter presented the experimental validation and performance evaluation of the Smart Real-Time Occupancy Monitoring System. The system was tested under various movement conditions to verify its accuracy, response time, and reliability.

The hardware testing confirmed that the IR sensors and processing logic accurately detected entry and exit movements and updated the occupancy count correctly. The system maintained stable performance during continuous operation with minimal delay.

The results demonstrate that the proposed system meets its objectives of providing real-time, accurate, and efficient occupancy monitoring. The project successfully proves that a hardware-based approach can offer reliable performance for safety and monitoring applications.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

6.1 Conclusion

The Smart Real-Time Occupancy Monitoring System successfully demonstrates a hardware-based approach for accurate and efficient people counting. The system reliably detects entry and exit movements and updates the occupancy count in real time with high accuracy and minimal delay. The use of hardware logic ensures fast response, low power consumption, and consistent performance compared to software-based solutions. The system effectively addresses issues such as overcrowding and inaccurate manual counting, making it suitable for safety-critical environments. Overall, the project validates that a hardware-based occupancy monitoring system can provide a reliable, scalable, and cost-effective solution for real-time monitoring applications.

6.2 Future Scope

Although the system achieves its primary objectives, there is scope for further enhancement:

- Integration with IoT platforms for remote monitoring and data analytics
- Implementation of wireless communication for centralized control
- Enhancement using advanced sensors or vision-based systems for higher accuracy
- Miniaturization of hardware for compact and easy deployment
- Extension to large-scale environments with multi-point monitoring systems

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SDG and TRL Mapping

PROJECT TITLE	Smart Room Occupancy Monitoring System
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1. Sustainable Development Goals (SDG) Mapping and Justification

S.No	SDG Number	SDG Title	Justification
1	SDG 9	Industry, Innovation and Infrastructure	Enhances smart infrastructure by providing an innovative real-time occupancy monitoring system.
2	SDG 11	Sustainable Cities and Communities	Enhances safety in public spaces by preventing overcrowding through real-time occupancy monitoring.

2. Technology Readiness Level (TRL) Mapping and Justification

TRL Level	Description	Justification
TRL 5	System Validation in Relevant Environment	• A functional prototype of the Smart Real-Time Occupancy Monitoring System was developed using IR sensors and hardware logic. • The system was tested in a controlled indoor environment by simulating entry and exit movements. • The results confirmed accurate bidirectional counting and real-time response with minimal delay. • The system demonstrated reliable performance in a relevant environment, validating the feasibility of the design.

Signature of the Supervisor
(Dr.K.N.Vijeyakumar, M.E, Ph.D.)



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PROJECT TITLE	Smart Room Occupancy Monitoring System	
PROGRAM	B.E. ELECTRONICS AND COMMUNICATION ENGINEERING	
PROJECT BATCH NUMBER	8	
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	722824106106	Nithishwaran S
	722824106121	Pravin A
	722824106129	Roobak S
NAME OF THE SUPERVISOR	Dr. K.N. Vijeyakumar, M.E., Ph.D.	
NAME OF THE SDG GOALS MAPPED	Industry, Innovation, and Infrastructure. Sustainable Cities and Communities.	
MENTION THE SDG GOALS NUMBER	9, 11	
NAME OF THE TRL LEVEL	Technology Validated in Relevant laboratory Environment	
MENTION THE TRL LEVEL	5	

POs & PSOs Mapping (Put a tick mark in the mapped PO's & PSO's):

Program Outcomes											Program Specific Outcomes	
PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
✓	✓	✓	✓	✓	✓			✓	✓	✓	✓	✓

Signature of the Supervisor
(Dr.K.N.Vijeyakumar, M.E., Ph.D.)



Sri Eshwar
College of Engineering
Coimbatore | Tamilnadu
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VENUE AND EXPENDITURE STATEMENT FOR THE PROJECT WORK

Laboratory details where the project is carried out	Project Lab, ECE
Software / Hardware details	Hardware: IR sensors (for entry/exit detection), FPGA development board (PYNQ-Z2), power supply module, connecting wires and interfacing components. Software: Verilog HDL (for hardware design), Vivado Design Suite (for synthesis and implementation testbench verification), PYNQ framework (for FPGA configuration and testing), Cadence (ASIC implementation)

Details of the Component and Expenditure

S. No	Name of the Component	Qty	Price / Unit in (Rs.)	Amount (Rs.)
1	PYNQ-Z2 FPGA Development Board	1	9000	9000
2	IR Sensor Module	2	80	160
3	Power Supply Module	1	200	200
4	Connecting Wires	1	100	100
			TOTAL	9460

Signature of the student

Laksana A

Nithishwaran S

Pravin A

Roobak S

Signature of the Supervisor

(Dr. K.N. Vijeyakumar, M.E., Ph.D.)